Set 1: MACHINE HEALTH & MAINTENANCE

SQL Question 3

Write queries to return: (a) daily average Temperature, Vibration, Pressure by Plant, and (b) top 5 MachineID per Plant by total DefectCount.

Created a Database named **Machine_health.**

Query - CREATE DATABASE Machine_health;

Solution:

-- Query to return (a) daily average Temperature, Vibration, Pressure by Plant

```
SELECT
   Timestamp AS date,
    Plant,
   ROUND(AVG(Temperature)) AS avg_temperature,
   ROUND(AVG(Vibration)) AS avg_vibration,
   ROUND(AVG(Pressure))  AS avg_pressure
FROM Machine_health_data
GROUP BY
   Timestamp, Plant
ORDER BY
   Timestamp DESC, Plant ASC;
```

Output of the 3(a) Query….

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

date	Plant	avg_temperature	avg_vibration	avg_pressure
9/9/2025	Plant_A	64	5	28
9/9/2025	Plant_B	78	5	29
9/9/2025	Plant_C	70	3	32
9/8/2026	Plant_A	69	5	29
9/8/2026	Plant_B	70	4	31
9/8/2026	Plant_C	76	5	28
9/8/2025	Plant_A	66	5	30
9/8/2025	Plant_B	68	5	28
9/8/2025	Plant_C	66	4	28
9/7/2026	Plant_A	77	6	30
9/7/2026	Plant_B	69	5	29
9/7/2026	Plant_C	71	5	31
9/7/2025	Plant_A	71	4	28
9/7/2025	Plant_B	72	5	29
9/7/2025	Plant_C	63	6	31
9/6/2026	Plant_A	73	5	30
9/6/2026	Plant_B	71	5	31
9/6/2026	Plant_C	68	6	31
9/6/2025	Plant_A	72	5	30
9/6/2025	Plant_B	71	4	28
9/6/2025	Plant_C	74	5	32
9/5/2026	Plant_A	69	5	27
9/5/2026	Plant_B	74	5	33
9/5/2026	Plant_C	70	6	31
9/5/2025	Plant_A	77	5	30
9/5/2025	Plant_B	68	5	28
9/5/2025	Plant_C	69	5	31

Daily average temperature, vibration, and pressure were calculated per plant using grouped timestamp data.

```sql
-- Query to return  (b) top 5 MachineID per Plant by total DefectCount

(SELECT
 Plant,
 MachineID,
 SUM(DefectCount) AS total_defects
FROM machine_health_data
WHERE Plant = 'Plant_A'
GROUP BY MachineID
ORDER BY SUM(DefectCount) DESC
LIMIT 5)

UNION ALL

(SELECT
 Plant,
 MachineID,
 SUM(DefectCount) AS total_defects
FROM machine_health_data
WHERE Plant = 'Plant_B'
GROUP BY MachineID
ORDER BY SUM(DefectCount) DESC
LIMIT 5)

UNION ALL

(SELECT
 Plant,
 MachineID,
 SUM(DefectCount) AS total_defects
FROM machine_health_data
WHERE Plant = 'Plant_C'
GROUP BY MachineID
ORDER BY SUM(DefectCount) DESC
LIMIT 5);
```

Output of the 3(b) Query….

Result Grid | Filter Rows:

Plant	MachineID	total_defects
Plant_A	139	541
Plant_A	137	494
Plant_A	125	474
Plant_A	132	471
Plant_A	121	471
Plant_B	115	531
Plant_B	103	498
Plant_B	118	495
Plant_B	126	485
Plant_B	106	480
Plant_C	119	491
Plant_C	100	490
Plant_C	129	483
Plant_C	132	475
Plant_C	105	473

Top 5 machines with highest total defect counts were identified per plant using UNION ALL.

SQL Question 4

Create a view machine_risk_2025 that assigns RiskLevel per MachineID using thresholds derived from dataset percentiles: High risk if Vibration >= 90th percentile OR Temperature >= 90th percentile OR DefectCount >= 90th percentile (else Medium if >= 75th any, else Low).

```
CREATE VIEW machines_risks_2025 AS
SELECT
    Timestamp,
    MachineID,
    Temperature,
    Vibration,
    DefectCount,

    CASE
        WHEN PERCENT_RANK() OVER (ORDER BY Vibration) >= 0.90
          OR PERCENT_RANK() OVER (ORDER BY Temperature) >= 0.90
          OR PERCENT_RANK() OVER (ORDER BY DefectCount) >= 0.90
        THEN 'High'
        WHEN PERCENT_RANK() OVER (ORDER BY Vibration) >= 0.75
          OR PERCENT_RANK() OVER (ORDER BY Temperature) >= 0.75
          OR PERCENT_RANK() OVER (ORDER BY DefectCount) >= 0.75
        THEN 'Medium'
        ELSE 'Low'
    END AS RiskLevel
FROM machine_health_data;

SELECT *FROM machines_risks_2025;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Timestamp	MachineID	Temperature	Vibration	DefectCount	RiskLevel
1/8/2027	120	29.8953902061528	4.57224091550704	4	Low
12/20/2025	105	33.9024300131197	4.69421319713656	6	High
12/25/2026	123	33.9537971231007	6.17276421144113	3	Medium
12/19/2025	114	35.4211670460228	3.19226841793888	1	Low
2/24/2026	100	35.7790873790111	5.93795991105955	2	Low
5/15/2025	115	35.8784404252085	7.4138667433034	0	High
4/9/2025	104	36.1676292754334	2.93885865359736	3	Low
2/24/2026	113	36.3804290036224	6.43865921079992	3	Medium
7/31/2026	143	36.675486166565	6.55506395910494	4	Medium
12/17/2025	114	37.1757012638725	5.48955206614801	2	Low
2/28/2026	110	37.2162745001257	5.4987248587304	3	Low
6/8/2026	112	37.454712998809	5.34008449381478	2	Low
1/30/2025	107	37.527250638064	2.59312328060921	3	Low
5/21/2025	100	38.6657903832665	2.98312010839806	8	High
6/26/2025	137	38.6841876585446	3.24013314874085	2	Low
9/26/2026	148	38.8452942081067	4.31485659633592	3	Low
1/2/2025	123	39.2086871615143	5.07362136964641	3	Low
1/19/2025	106	39.2512908387382	5.38873445808067	2	Low
8/28/2025	115	39.274870779086 6	3.08599448355871	5	Medium
3/25/2026	138	39.4927142831768	8.63162765347565	2	High
9/15/2026	148	39.6269113198839	5.42176956020395	2	Low
11/17/2025	133	39.6623741086802	3.42285184194733	2	Low
11/25/2026	112	39.9889823928552	4.55106788399584	2	Low
11/22/2025	138	40.2010387141548	3.94910102290313	5	Medium
1/1/2026	115	40.2701851651361	7.44802533375299	4	High
2/26/2026	133	40.4479838244712	4.60418492936817	5	Medium

Machines classified as High, Medium, or Low risk based on sensor and defect percentile thresholds.